

PAD21P03L – PROJECT WORK - FINAL REVIEW

KNEE OSTEOARTHRITIS DETECTION (KL-GRADE I – IV)

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ABSTRACT

- Knee Osteoarthritis (OA) is one of the most common degenerative joint diseases that affects millions of people worldwide, leading to pain, stiffness, and reduced mobility.
- The proposed system utilizes convolutional neural network (CNN) architectures to analyze X-ray images and classify them into different stages of osteoarthritis based on the Kellgren–Lawrence (KL) grading scale.
- The trained model learns important patterns in X-ray images that correspond to structural changes in the knee joint, enabling accurate classification of OA severity levels.
- The application displays the predicted KL grade, prediction confidence, class probability distribution, and a Grad-CAM heatmap that highlights the regions of the X-ray image influencing the model’s decision.
- The developed system demonstrates that deep learning techniques can effectively assist in the automated detection and grading of knee osteoarthritis.
- By providing fast, reliable, and interpretable predictions, the proposed solution can support medical professionals in early diagnosis and clinical decision-making, ultimately improving patient care and reducing the burden on healthcare systems.

PROBLEM STATEMENT

- Knee Osteoarthritis (OA) is a progressive joint disorder that significantly affects the mobility and quality of life of individuals, especially among the elderly population.
- The manual interpretation process can be time-consuming and may lead to inconsistencies due to subjective judgment.
- In many healthcare settings, the growing number of patients and the limited availability of medical specialists make it difficult to perform quick and accurate diagnosis for every case.
- As a result, there is a need for an automated and reliable system that can assist medical professionals in detecting and classifying the severity of knee osteoarthritis from X-ray images.
- Therefore, the main problem addressed in this project is the development of an intelligent deep learning–based system capable of automatically analyzing knee X-ray images and accurately classifying them into different osteoarthritis severity levels according to the KL grading scale.
- By doing so, the proposed solution aims to support medical professionals in improving diagnostic accuracy, reducing analysis time, and enabling faster clinical decision-making.

FEASIBILITY STUDY

- The proposed system is technically feasible because it uses widely available technologies such as Python, deep learning frameworks, and web-based interfaces.
- Since these technologies are well-supported and can run on standard computers with GPU or cloud support, the technical implementation of the system is practical and achievable.
- The system is designed to be user-friendly and easy to operate for both medical professionals and general users.
- Because the system simplifies the diagnosis process and requires minimal technical knowledge to operate, it can easily be integrated into clinical workflows or used as a supportive diagnostic tool.
- The proposed system is economically feasible because it primarily relies on open-source technologies and freely available tools such as Python, TensorFlow, and Streamlit. This significantly reduces development and operational costs.
- By automating the detection process, the system can also reduce the workload on radiologists and improve diagnostic efficiency, ultimately lowering the overall cost associated with manual diagnosis and repeated medical examinations.

EXISTING SYSTEM

- In the existing system, the diagnosis of knee osteoarthritis is mainly performed through manual analysis of knee X-ray images by experienced radiologists or orthopedic specialists.
- The severity of the disease is determined using the Kellgren–Lawrence (KL) grading system, which classifies osteoarthritis into different stages based on visible structural changes such as joint space narrowing, osteophyte formation, and bone sclerosis.
- Medical experts visually inspect the X-ray images and assign a grade according to their interpretation of these features.
- Although this traditional method has been widely used in medical practice, it relies heavily on human expertise and experience.
- Additionally, analyzing large numbers of X-ray images manually can be time-consuming and may delay the diagnosis and treatment process, especially in healthcare facilities with limited medical staff.

DRAWBACKS OF EXISTING SYSTEM

- One of the major drawbacks of the existing system is the dependence on manual interpretation by radiologists. Since the diagnosis relies on human observation, there is a possibility of subjective variation between experts.
- Another limitation is that the manual analysis of X-ray images requires significant time and effort, especially when dealing with a large number of patients.
- In busy hospitals or medical centers, this can increase the workload on healthcare professionals and may lead to delays in providing accurate diagnostic results to patients.
- Furthermore, early stages of osteoarthritis often show very subtle changes in the knee joint that may not be easily visible through visual inspection alone.
- As a result, the disease may only be diagnosed at a later stage when the condition has already progressed, reducing the effectiveness of early treatment and preventive measures.

PROPOSED SYSTEM

- The proposed system introduces an automated deep learning–based approach for detecting and classifying knee osteoarthritis using knee X-ray images.
- These models can accurately classify the severity of osteoarthritis into multiple categories, enabling faster and more consistent diagnosis.
- In addition to automated classification, the system includes a user-friendly web application developed using Streamlit.
- Users can upload knee X-ray images through the interface, and the system will analyze the image and display the predicted KL grade along with prediction confidence and probability distribution.
- The system also provides visualization using Grad-CAM heatmaps, which highlight the regions of the X-ray image that influenced the model’s prediction, making the results more interpretable.

BENEFITS OF PROPOSED SYSTEM

- The proposed project offers several advantages, including faster and more accurate detection of knee osteoarthritis.
- By automating the analysis of X-ray images, the system reduces the workload on medical professionals and minimizes the chances of human error in diagnosis.
- The use of deep learning models improves the ability to detect subtle patterns in medical images, enabling earlier identification of osteoarthritis.
- Additionally, the interactive web application provides an easy-to-use platform that allows users to obtain quick predictions and visual explanations, making the system useful as a supportive diagnostic tool in healthcare environments.

SCOPE OF THE PROJECT

- The scope of this project focuses on developing an intelligent system capable of detecting and classifying knee osteoarthritis severity using X-ray images.
- The system can be used as a supportive diagnostic tool for healthcare professionals, assisting them in evaluating the progression of osteoarthritis more efficiently.
- In the future, the system can be expanded by training with larger medical datasets, improving model accuracy, and integrating additional medical imaging techniques such as MRI or CT scans.
- It may also be deployed in hospital systems or telemedicine platforms to provide remote diagnostic support and improve accessibility to healthcare services.

HARWARE REQUIREMENTS

- System Laptop
- Device name DESKTOP-HLLGJE2
- Processor 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz (2.42 GHz)
- Installed RAM 16.0 GB (15.7 GB usable)
- Device ID 1B2D7FB8-5561-4E26-9F31-9AE808FAC1EB
- Product ID 00326-10000-00000-AA668
- System type 64-bit operating system, x64-based processor
- Pen and touch No pen or touch input is available for this display

SOFTWARE REQUIREMENTS

Category

Programming Language

Development Environment

Deep Learning Framework

Machine Learning Libraries

Image Processing Libraries

Data Handling Libraries

Visualization Libraries

Web Application Framework

Software / Tools Used

Python 3.8 / 3.9/3.10

Jupyter Notebook, Google Colab, KaggleNotebook, VS Code

TensorFlow 2.x / Keras or PyTorch

Scikit-learn, Imbalanced-learn

OpenCV, PIL, Albumentations

NumPy, Pandas

Matplotlib, Seaborn, Grad-CAM / tf-keras-vis

Streamlit

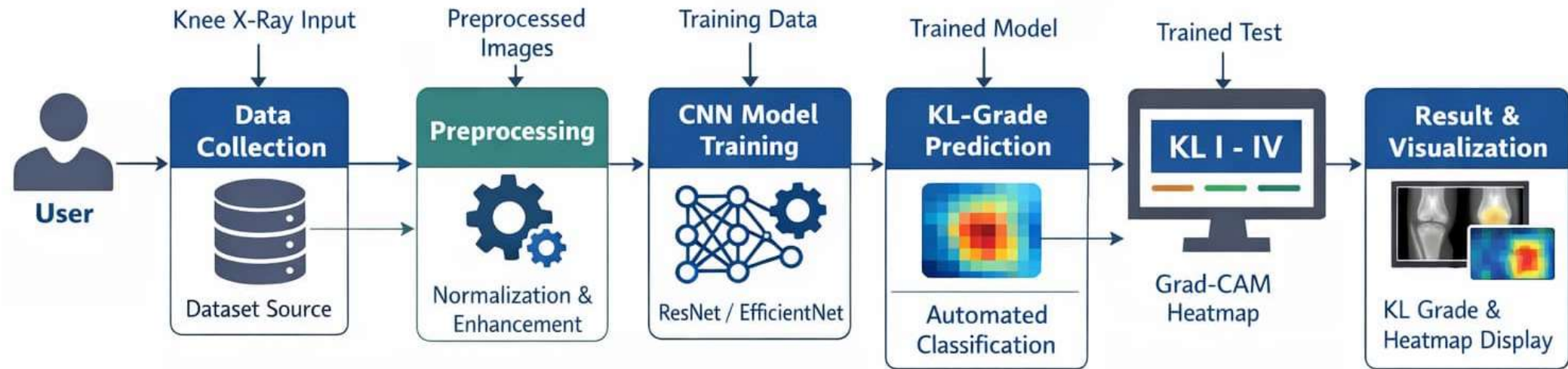
MODULE DESCRIPTION

- **Data Collection Module:** This module handles the acquisition of knee X-ray images from publicly available datasets with KL-grade annotations (Grade I–IV). It ensures proper dataset organization and label mapping for training and evaluation.
- **Data Preprocessing Module:** This module performs image resizing, normalization, noise removal, contrast enhancement, and data augmentation. These steps improve image quality and help the model learn relevant features effectively.
- **Dataset Balancing Module:** This module addresses class imbalance using techniques such as oversampling, undersampling, or weighted loss functions to ensure fair learning across all KL grades.
- **Model Development Module:** This module implements deep learning architectures such as ResNet or EfficientNet. It includes model configuration, training, and fine-tuning for optimal performance in KL-grade classification.

MODULE DESCRIPTION

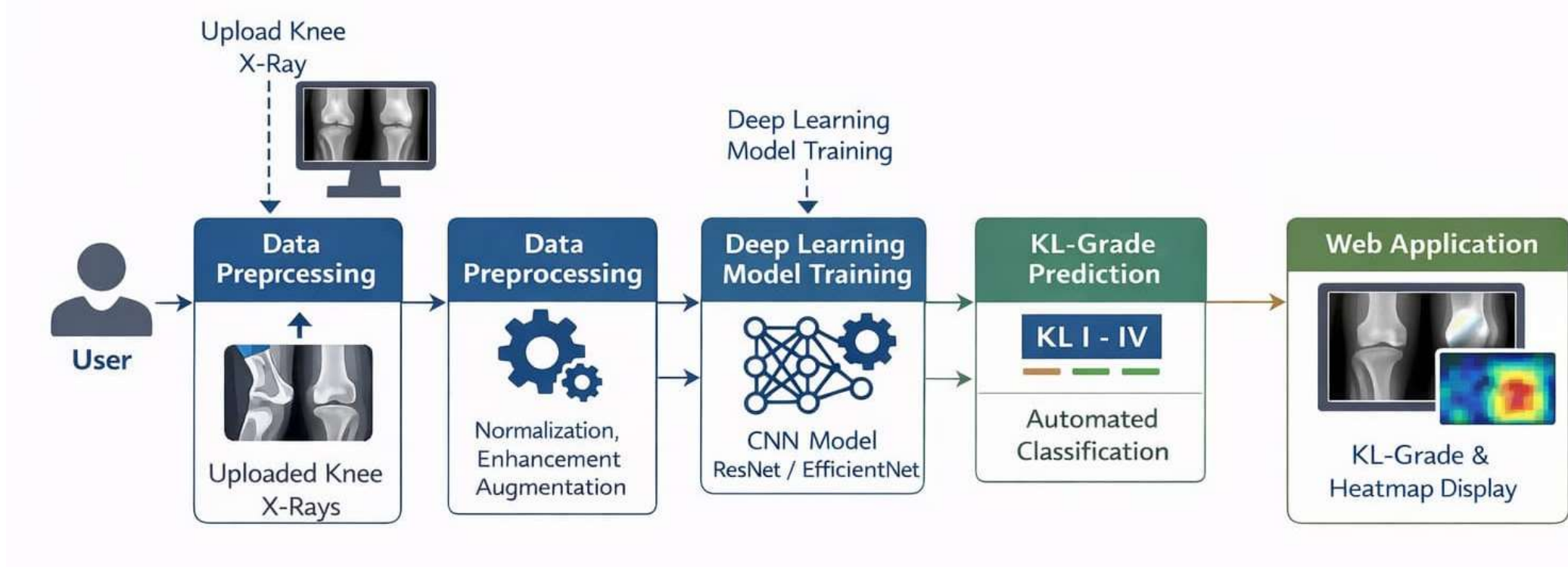
- **Model Evaluation Module:** This module evaluates model performance using accuracy, precision, recall, F1-score, and confusion matrix to assess classification effectiveness.
- **Prediction Module:** This module accepts new or external knee X-ray images and generates KL-grade predictions in real time using the trained model.
- **Explainability Module (Grad-CAM):** This module generates Grad-CAM heatmaps to highlight regions of interest in knee X-rays, providing visual explanations that support clinical interpretation.
- **Web Application Module:** This module deploys the trained model using Streamlit, enabling users to upload X-ray images and view predictions and heatmaps through an interactive interface.
- **Model Storage and Management Module:** This module manages trained model saving, loading, and version control to support reuse and future enhancements.

DATAFLOW DIAGRAM



- The Dataflow Diagram explains the process: from input X-rays through preprocessing, CNN model training, KL-grade prediction explainability to displaying results.

ARCHITECTURE DIAGRAM



- The Architecture Diagram explains the process: from uploading knee X-rays, through preprocessing, CNN model training. KL-grade prediction, to displaying results using a web application.

DATASET DESCRIPTION

- The dataset used in this project consists of labeled knee X-ray images categorized according to the Kellgren–Lawrence grading system.
- Each image represents a specific stage of osteoarthritis ranging from Normal, Doubtful, Mild, Moderate, to Severe.
- The dataset is divided into training, validation, and testing subsets to ensure proper model learning and performance evaluation.
- Each image undergoes preprocessing steps such as resizing, normalization, and augmentation to improve model accuracy and prevent overfitting.
- The dataset provides sufficient variability in knee joint conditions, which helps the deep learning models learn meaningful patterns for accurate classification of osteoarthritis severity.

CODING SCREENSHOTS

```
model_eff, acc_eff, time_eff, h1_eff, h2_eff = train_model("EfficientNetB3", EfficientNetB3, eff_pre)
model_res, acc_res, time_res, h1_res, h2_res = train_model("ResNet50", ResNet50, res_pre)
model_dense, acc_dense, time_dense, h1_dense, h2_dense = train_model("DenseNet121", DenseNet121, dense
```

```
EfficientNetB3 Test Accuracy: 64.49%
Training Time: 55.93 minutes
```

```
===== ResNet50 =====
```

```
Found 5778 images belonging to 5 classes.
```

```
Class Weights: {0: np.float64(0.5055118110236221), 1: np.float64(1.1047801147227534), 2: np.float64(0.7622691292875989), 3: np.float64(1.5265521796565389), 4: np.float64(6.679768786127168)}
```

```
Found 826 images belonging to 5 classes.
```

```
Found 1656 images belonging to 5 classes.
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/trainers/data_adapters/py_dataset_adapter.py:12
1: UserWarning: Your `PyDataset` class should call `super().__init__(**kwargs)` in its constructor. `**kwargs` can include `workers`, `use_multiprocessing`, `max_queue_size`. Do not pass these arguments to `fit()`, as they will be ignored.
  self._warn_if_super_not_called()
```

```
Epoch 1/50
```

```
362/362 ————— 98s 241ms/step - accuracy: 0.2677 - loss: 0.6615 - val_accuracy: 0.38
```

```
86 - val_loss: 0.2717 - learning_rate: 0.0010
```

```
Epoch 2/50
```

CODING SCREENSHOTS

```
model_eff, acc_eff, time_eff, h1_eff, h2_eff = train_model("EfficientNetB3", EfficientNetB3, eff_pre)
model_res, acc_res, time_res, h1_res, h2_res = train_model("ResNet50", ResNet50, res_pre)
model_dense, acc_dense, time_dense, h1_dense, h2_dense = train_model("DenseNet121", DenseNet121, dense_pre)

368 - val_loss: 1.0579 - learning_rate: 9.0000e-06
Epoch 18/25
362/362 ————— 128s 353ms/step - accuracy: 0.6735 - loss: 1.0098 - val_accuracy: 0.6
356 - val_loss: 1.0611 - learning_rate: 2.7000e-06
Epoch 19/25
362/362 ————— 129s 358ms/step - accuracy: 0.6704 - loss: 1.0065 - val_accuracy: 0.6
344 - val_loss: 1.0609 - learning_rate: 2.7000e-06
Epoch 20/25
362/362 ————— 128s 352ms/step - accuracy: 0.6778 - loss: 0.9998 - val_accuracy: 0.6
320 - val_loss: 1.0614 - learning_rate: 8.1000e-07
Epoch 21/25
362/362 ————— 131s 361ms/step - accuracy: 0.6705 - loss: 1.0077 - val_accuracy: 0.6
295 - val_loss: 1.0611 - learning_rate: 8.1000e-07
Epoch 22/25
362/362 ————— 126s 349ms/step - accuracy: 0.6587 - loss: 1.0143 - val_accuracy: 0.6
356 - val_loss: 1.0597 - learning_rate: 2.4300e-07
104/104 ————— 19s 181ms/step - accuracy: 0.7762 - loss: 0.8674
DenseNet121 Test Accuracy: 64.55%
Training Time: 75.83 minutes
```

OUTPUT

```
# Your model accuracies
results = {
    "EfficientNetB3": acc_eff,
    "ResNet50": acc_res,
    "DenseNet121": acc_dense
}

# Extract model names and accuracies
models = list(results.keys())
accuracies = [v * 100 for v in results.values()]

# Find best model
best_model = max(results, key=results.get)

# Set colors (best model in green)
colors = ['skyblue' if m != best_model else 'green' for m in models]

# Plot bar chart
plt.figure(figsize=(8,5))
plt.bar(models, accuracies, color=colors)

plt.title("Comparative Performance of CNN Models")
plt.xlabel("Models")
plt.ylabel("Accuracy (%)")

# Show accuracy values on bars
for i, v in enumerate(accuracies):
    plt.text(i, v + 0.5, f"{v:.2f}%", ha='center')

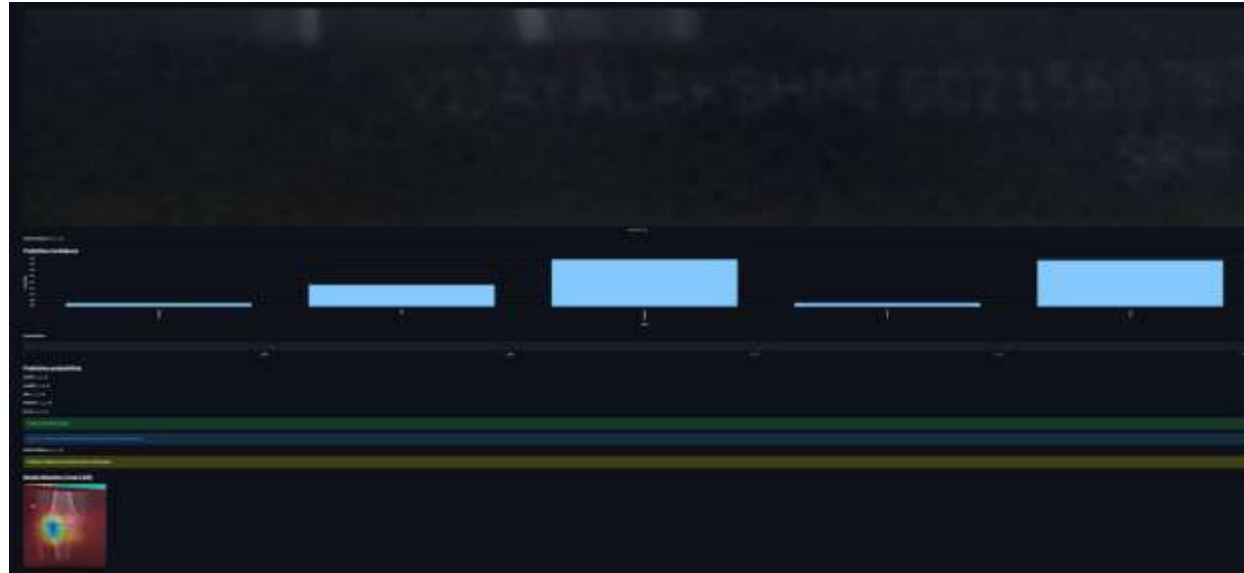
plt.ylim(0,100)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



INTERFACE



INTERFACE



SYSTEM TESTING

- **Functional Testing:** Functional testing ensures that every feature of the system works according to the specified requirements.
- **Usability Testing:** Usability testing evaluates how easily users can interact with the Streamlit web application.
- **Performance Testing:** Performance testing evaluates how efficiently the system processes images and generates predictions.
- **Compatibility Testing:** Compatibility testing ensures that the web application works properly across different systems and browsers.
- **Regression and Iterative Testing:** Regression testing ensures that updates or modifications to the system do not negatively affect existing features.

CONCLUSION

- The Knee Osteoarthritis Detection system based on KL grading (I–IV) successfully demonstrates the application of deep learning in medical image classification.
- The results confirm that deep learning models can reliably extract clinically relevant features such as joint space narrowing and bone deformities.
- The experimental results also highlight important dataset-related challenges.
- The combination of model training, evaluation metrics, and visualization techniques ensures both performance and transparency, making the system suitable for real-world assistance in clinical decision-making.
- However, further improvements in dataset quality and model tuning are required to achieve higher diagnostic precision, especially for early-stage detection.

FUTURE ENHANCEMENT

- The current dataset shows imbalance across KL grades, which affects model performance, especially for KL1.
- Although basic preprocessing is used, advanced augmentation techniques such as rotation, contrast enhancement, and noise injection can be applied.
- Further tuning of learning rate, batch size, optimizer selection, and fine-tuning layers can significantly improve performance.
- Future improvements can involve testing newer architectures such as Vision Transformers (ViTs) or hybrid CNN-transformer models.
- Techniques like SMOTE (Synthetic Minority Oversampling), focal loss, or balanced batch sampling can be applied to improve classification performance for minority classes, especially KL1.

THANK YOU